

**Topic 9: Organisms and their environment – Interactions between living organisms**

| Students should: |                                                                                                                                                   | Guidance<br><i>iLowerSecondary Curriculum reference</i>                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.1              | understand the terms <i>producer, primary consumer, secondary consumer, tertiary consumer</i> and <i>decomposer</i>                               | <i>Year 7: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.2              | know the interdependence of organisms in the environment, in terms of feeding relationships by drawing and interpreting food chains and food webs | use the terms <i>predator, prey, herbivore, carnivore</i> and <i>omnivore</i> correctly<br><br>be able to link photosynthesis to the producers in a food chain and also to global food supplies<br><br><i>Year 7: Organisms and their environment – Interactions between living organisms</i> |
| 9.3              | know how to draw and interpret pyramids of number                                                                                                 | <i>Year 8: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.4              | know how to interpret food chains and food webs in terms of energy flow                                                                           | <i>Year 8: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.5              | explain ways in which energy is lost between trophic levels                                                                                       | <i>Year 8: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.6              | know how to draw and interpret pyramids of biomass                                                                                                | <i>Year 9: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.7              | understand the terms <i>species, population, community, habitat</i> and <i>ecosystem</i>                                                          | <i>Year 9: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.8              | know the difference between inter- and intra-specific competition                                                                                 | <i>Year 9: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.9              | know how to interpret data on population numbers                                                                                                  | <i>Year 9: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.10             | explain why population numbers fluctuate, including predator-prey relationships                                                                   | <i>Year 9: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |
| 9.11             | know about predator-prey interactions in order to interpret data                                                                                  | <i>Year 9: Organisms and their environment – Interactions between living organisms</i>                                                                                                                                                                                                        |